Canary Deployment Using Lambda and API Gateway

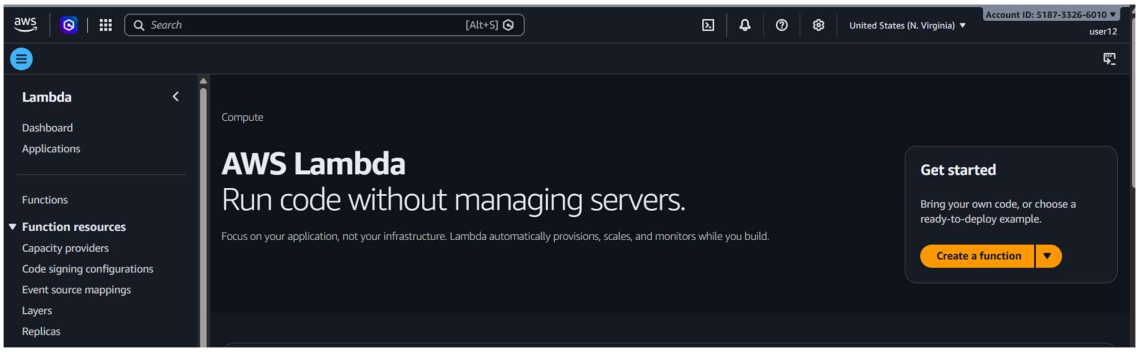
Problem Statement:

The ABC Company is continuously updating the application software. The application is deployed on Lambda and API Gateway. Recently new version of the application is being released. The CTO of the company wants to check the new version of the application with a sample set of traffic and if everything is working fine then a11100% of traffic will be redirected. As application developer you are instructed to use native feature of API Gateway for doing so.

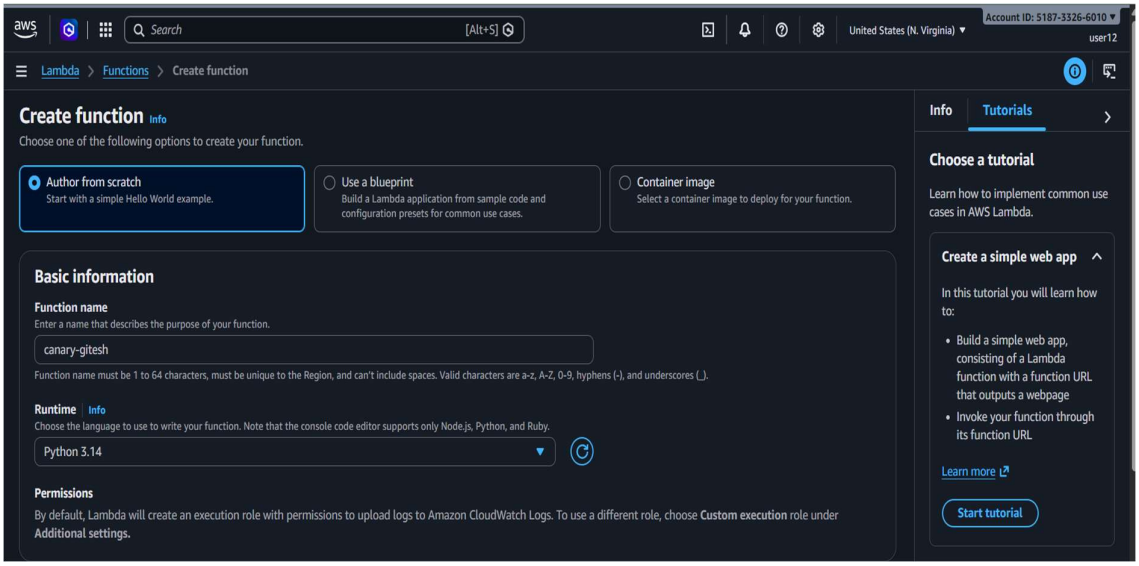
Solution :

Task 1 : Create a Lambda function

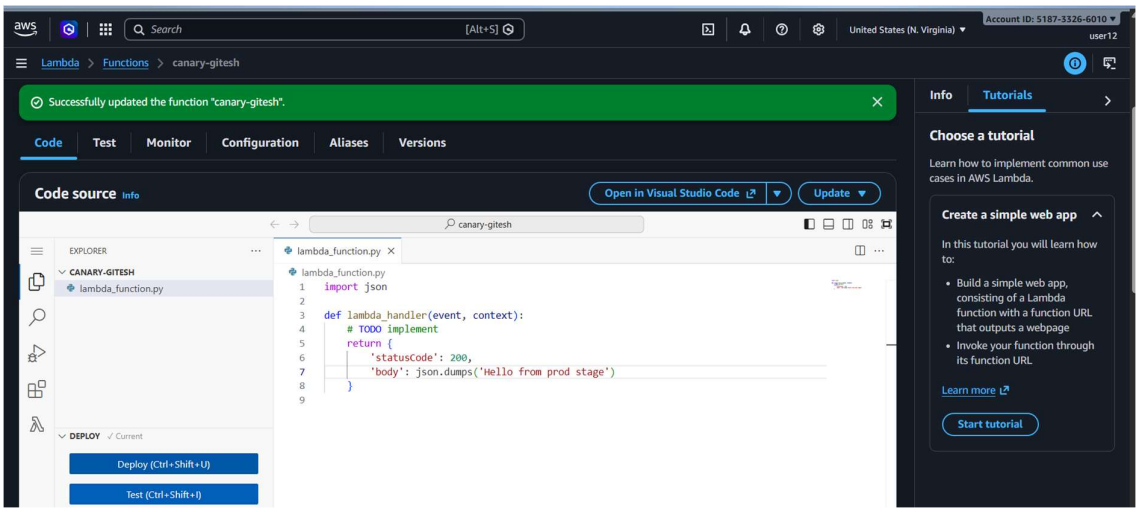
1. Go to Lambda Console
2. Choose Functions from Navigation Pane.
3. Select Create Function



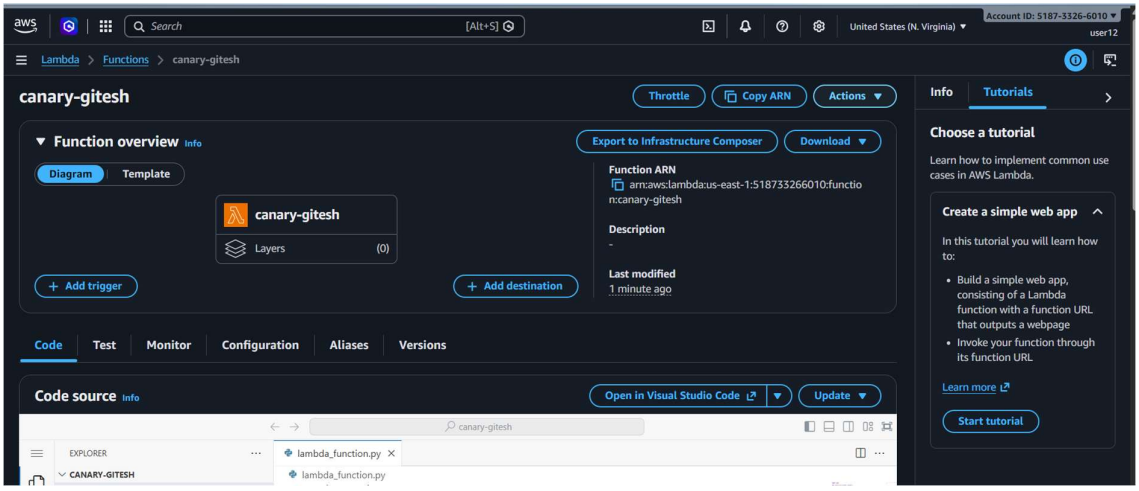
4. Choose Author from Scratch.
5. Enter canary-gitesh as function Name
6. For runtime, select Python 3.10 or latest Python Runtime.



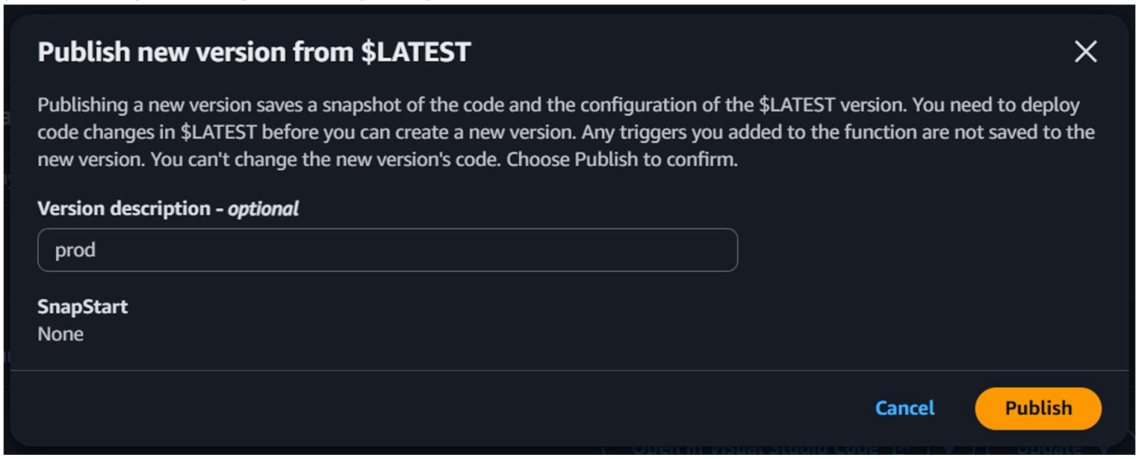
7. Select Create function.
8. Once the function is created, change the return text "Hello from prod stage" instead of "Hello from Lambda"
9. Deploy the changes.



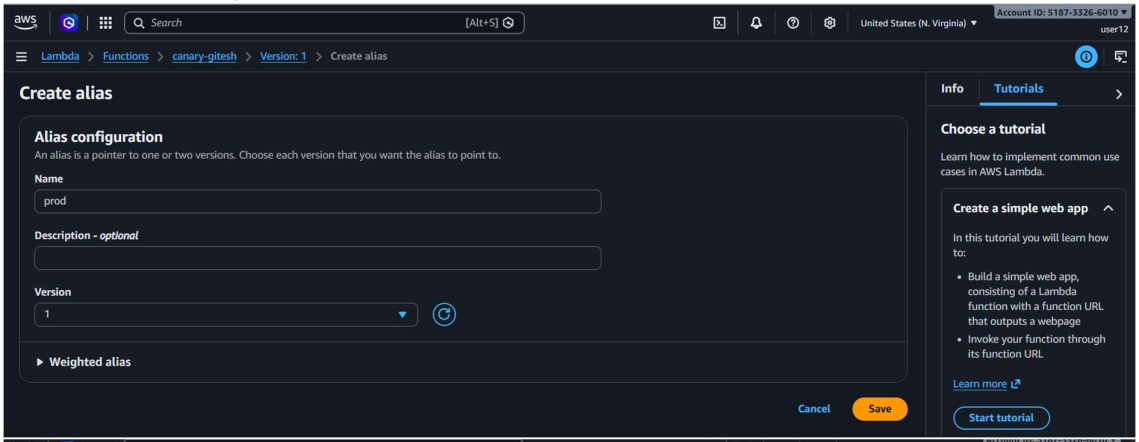
10. Go to action and select publish new version.

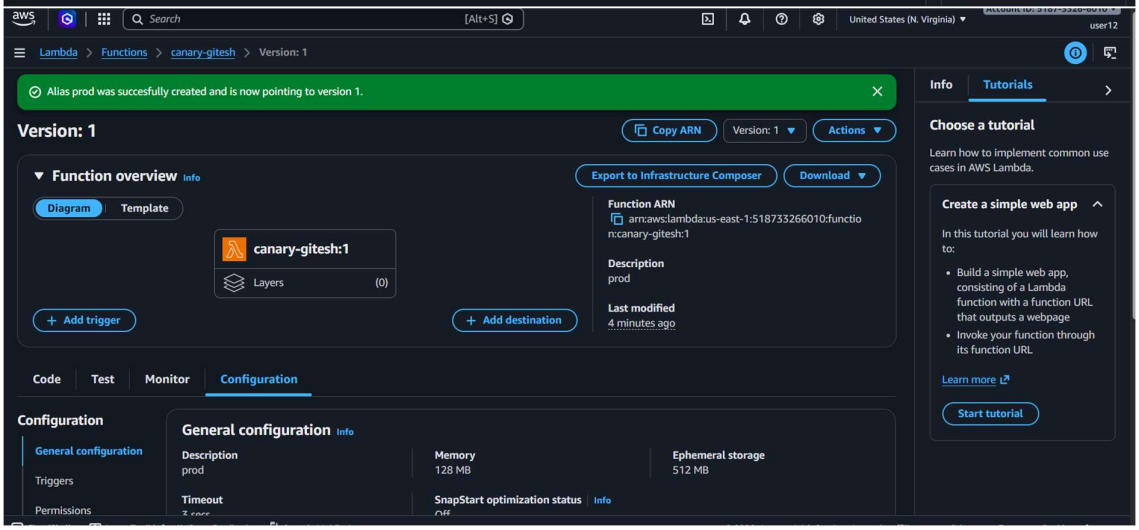


11. put a description as "prod" and press publish



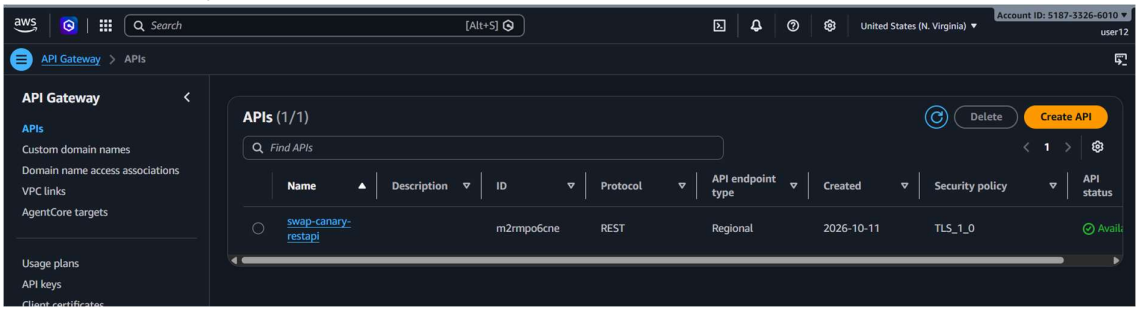
12. Go to action again, select Create alias
13. In alias configuration window , enter name of alias as prod
14. Select version 1 and press save button.



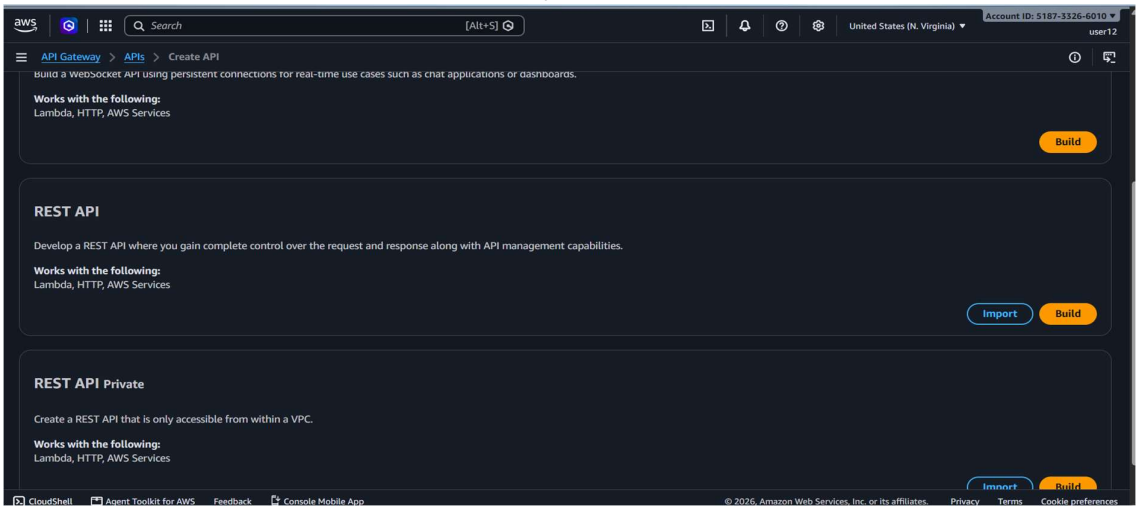


Task 2: Create an API Gateway

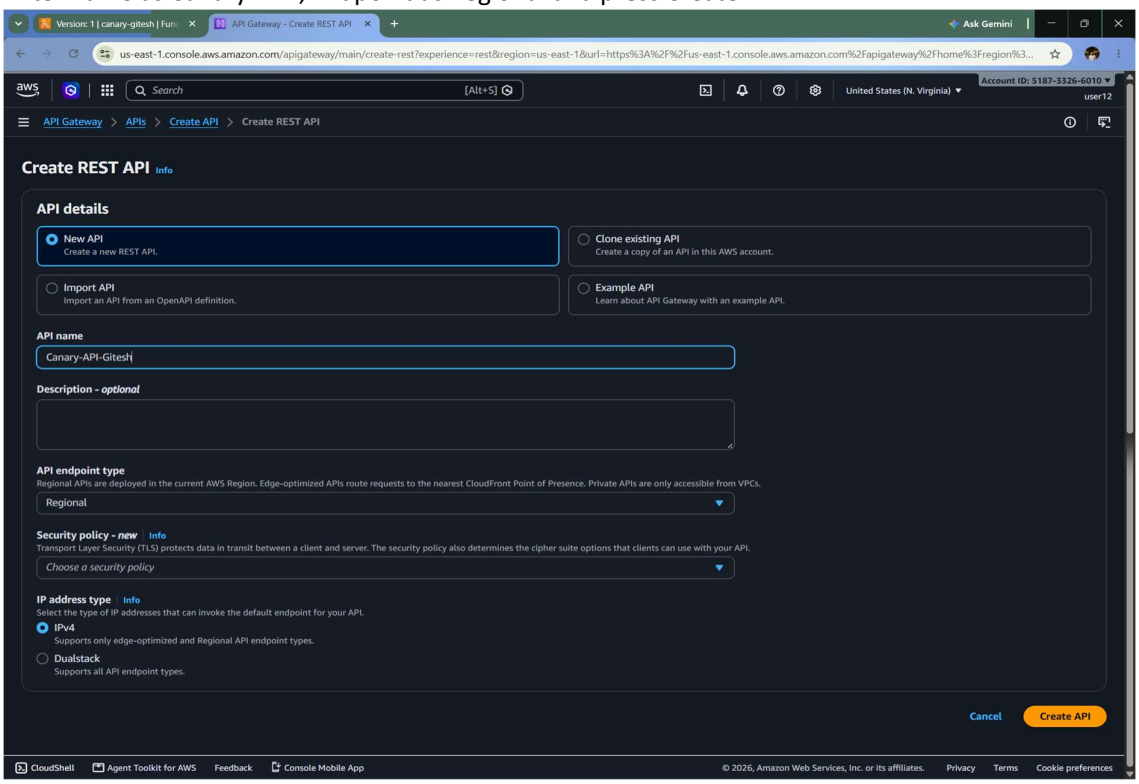
1. GO to API Gateway Console and select Create API.



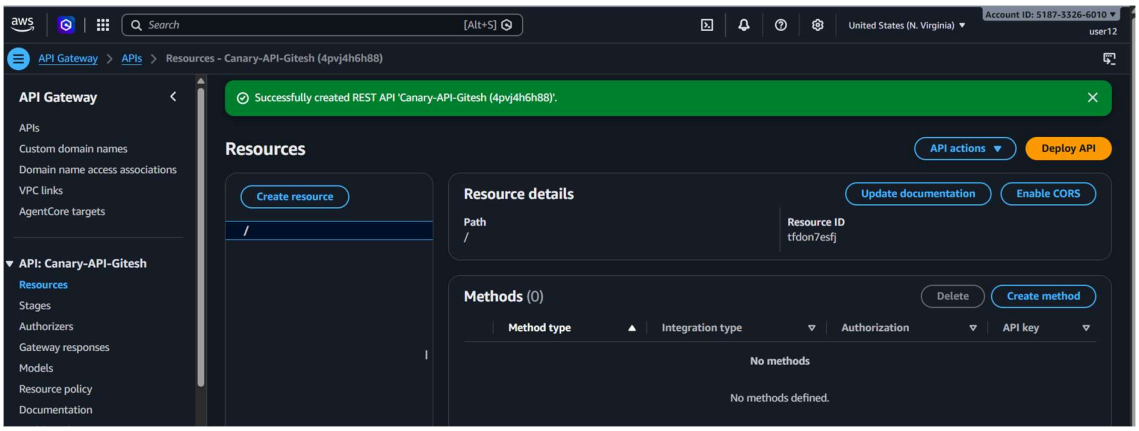
2. Under create API window select Rest based and press Build button



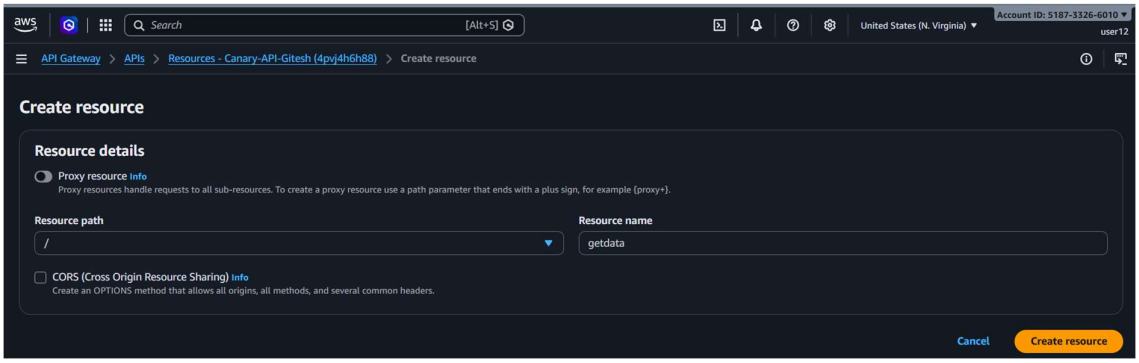
3. Enter name as Canary-API , Endpoint as Regional and press Create API.



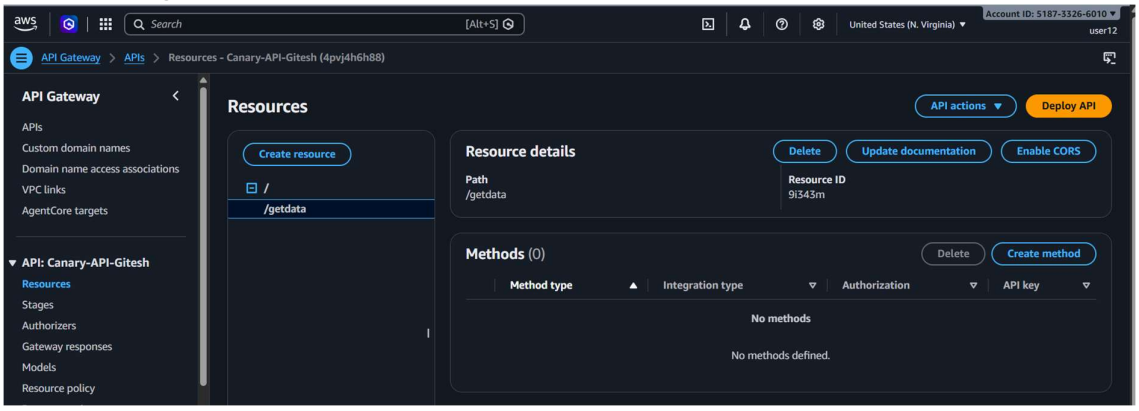
4. Go to action and create a resource



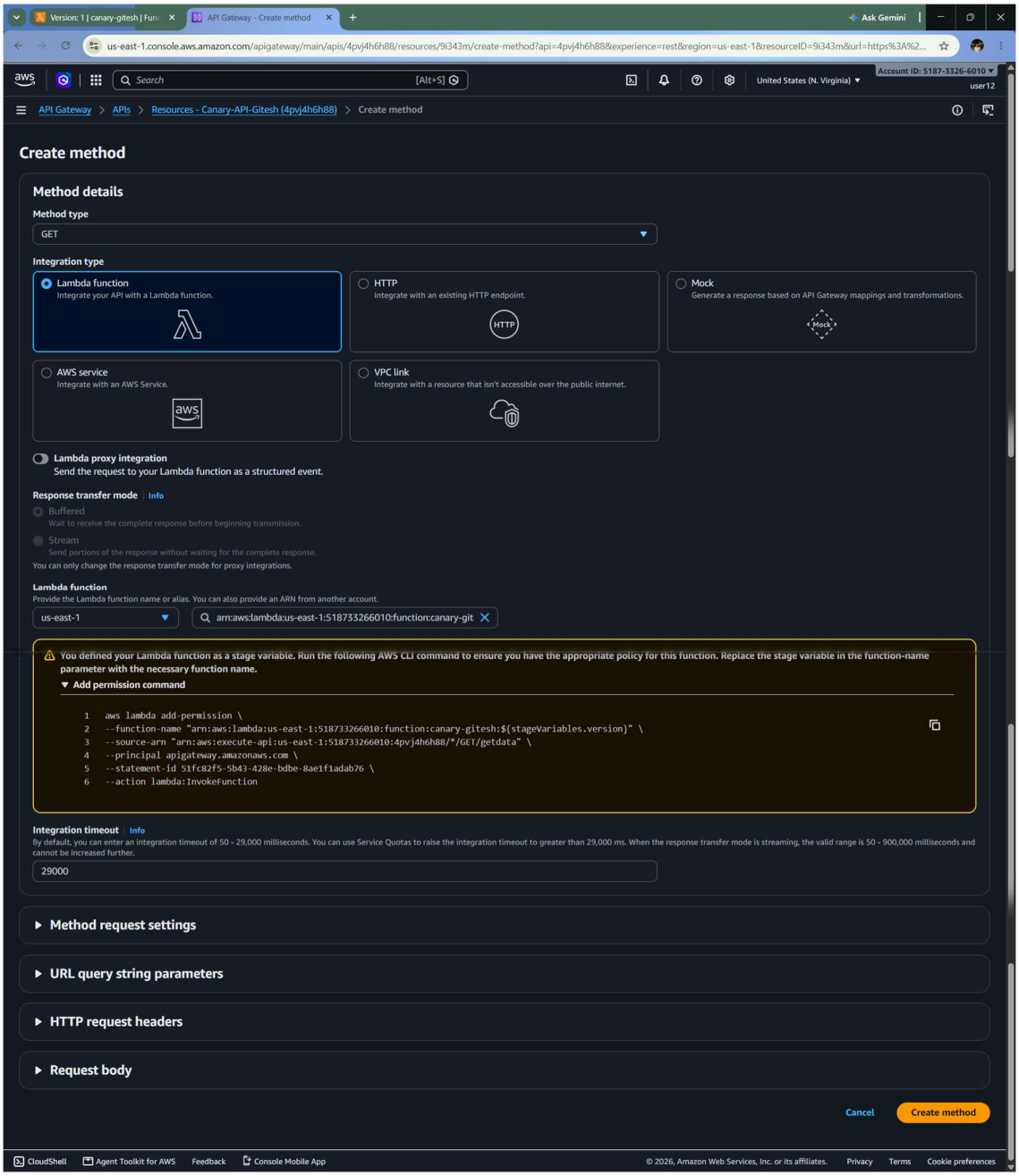
5. Under Resource name , enter resource name as getdata and press create Resource



6. Go to action again under resource , select create method and select GET from list



7. Select Get Method and press right arrow key
8. Now in Integration type select Lambda as Integration type.
9. In Lambda Function, Enter name of Lambda function followed by stage variable name as follows- canary-demo:${stagevariables.version} where version is name of stage variable and press save.
10. We will create Stage variable in task 2 step 18.

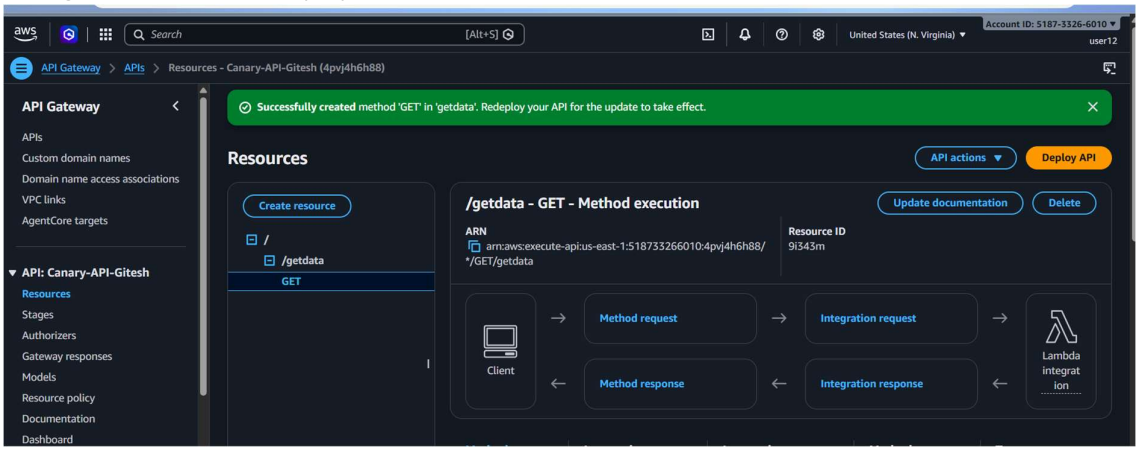


11. Once you press Save button it will ask you to add permissions
    aws lambda add-permission --function-name "arn:aws:lambda:us-east
    1:518733266010:function:canary-gitesh:1"
    --source-arn "arn:aws:execute-api:us-east-1:518733266010:4pvj4h6h88/*/GET/getdata"
    --principal apigateway.amazonaws.com
    --statement-id 51fc82f5-5b43-428e-bdbe-8ae1f1adab76
    --action lambda:InvokeFunction

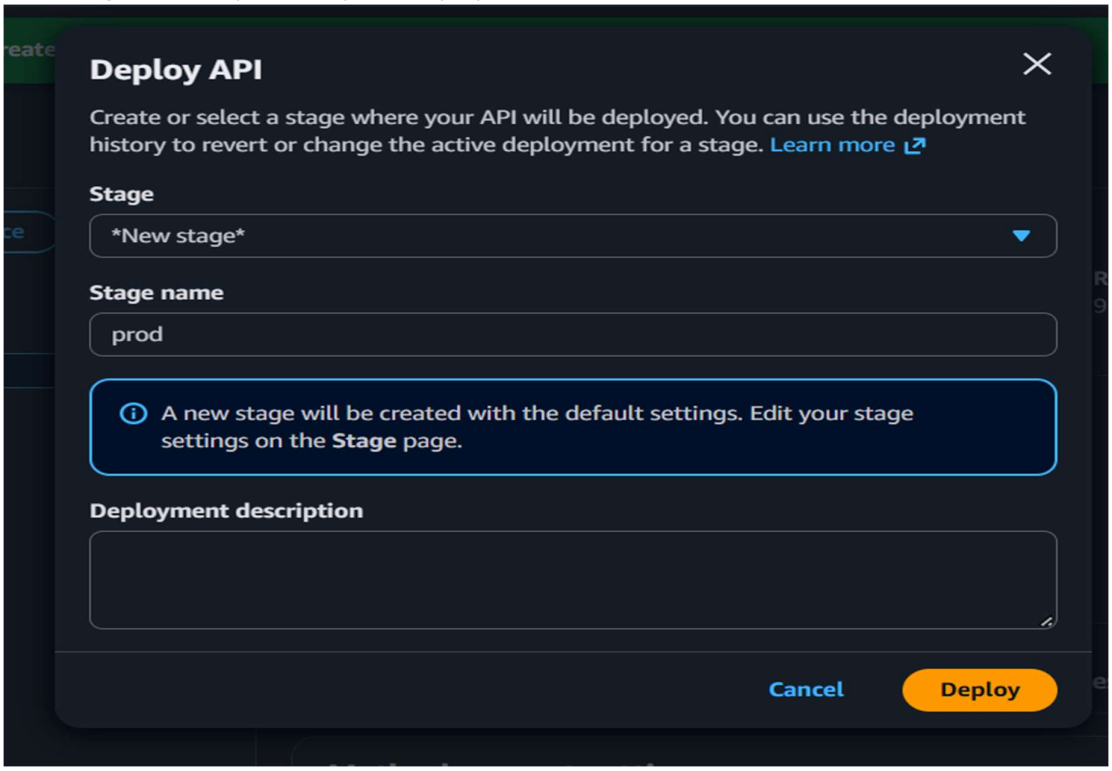
12. Copy and run those commands using AWS CLI.
13. Replace name Of alias with Stage variable in command as follows
14. Once added permissions click on OK



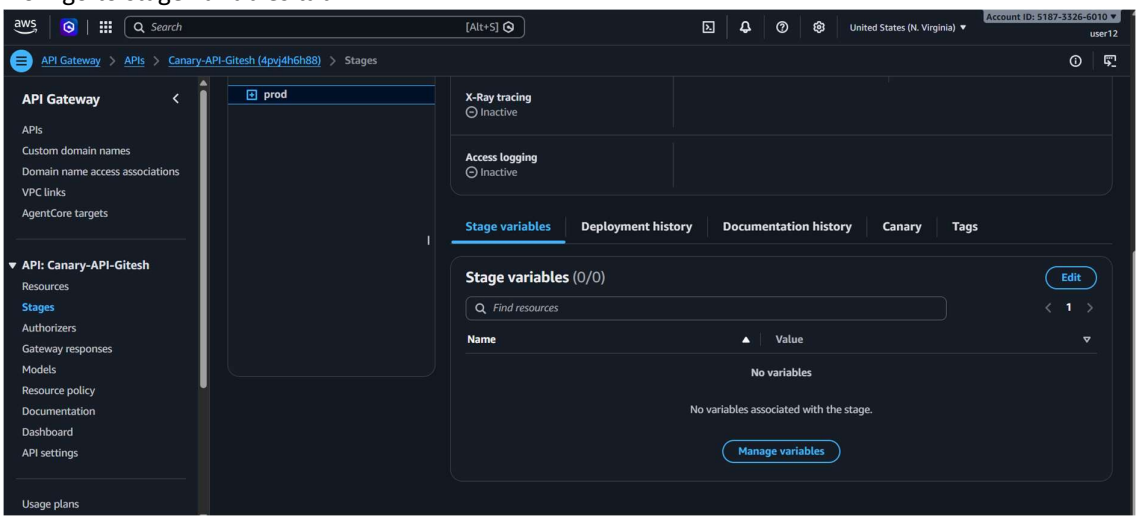
15. Now go to action select Deploy API



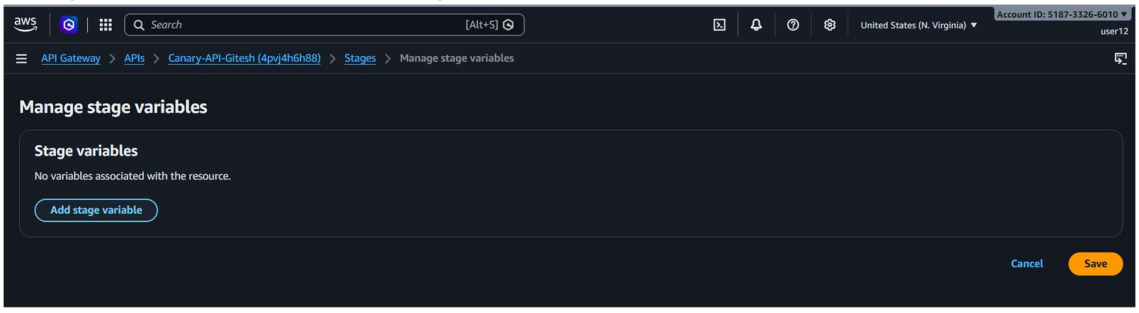
16. Under Deploy API window, select new Stage
17. Enter stage name as prod and press Deploy



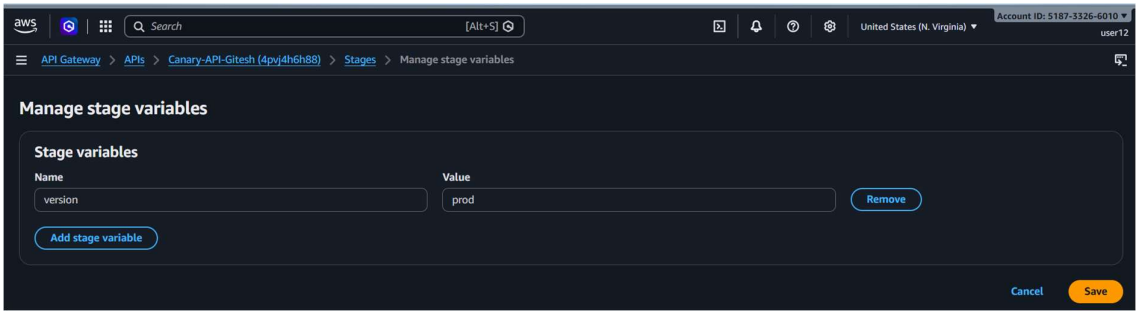
18. Now go to stage variables tab.



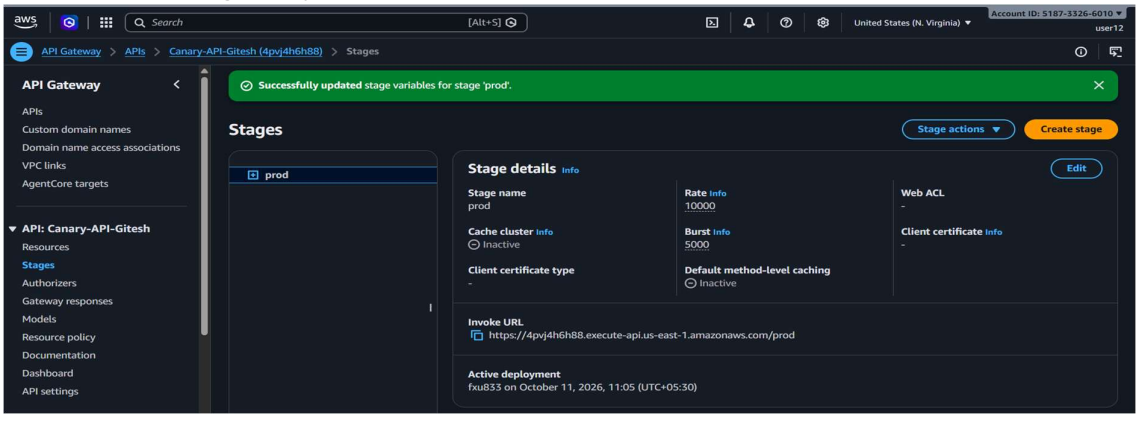
19. In Stage variables tab , click on add Stage variable



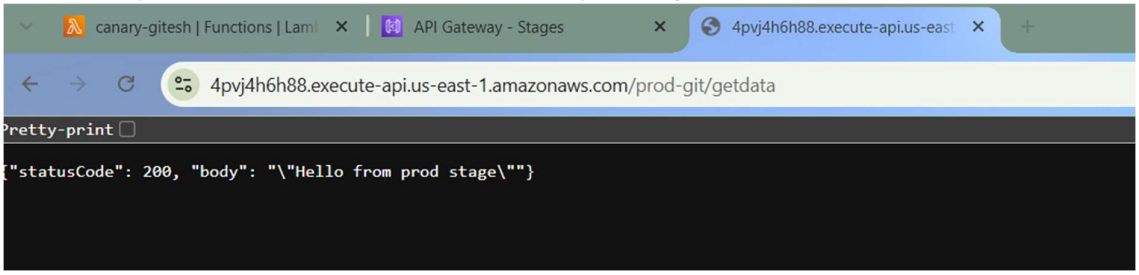
20. Under name enter stage variable name as version (as in Task 2 step 9 we have specified name as version) and value is name Of alias(in our case alias name is prod)
21. Press tick at the end.



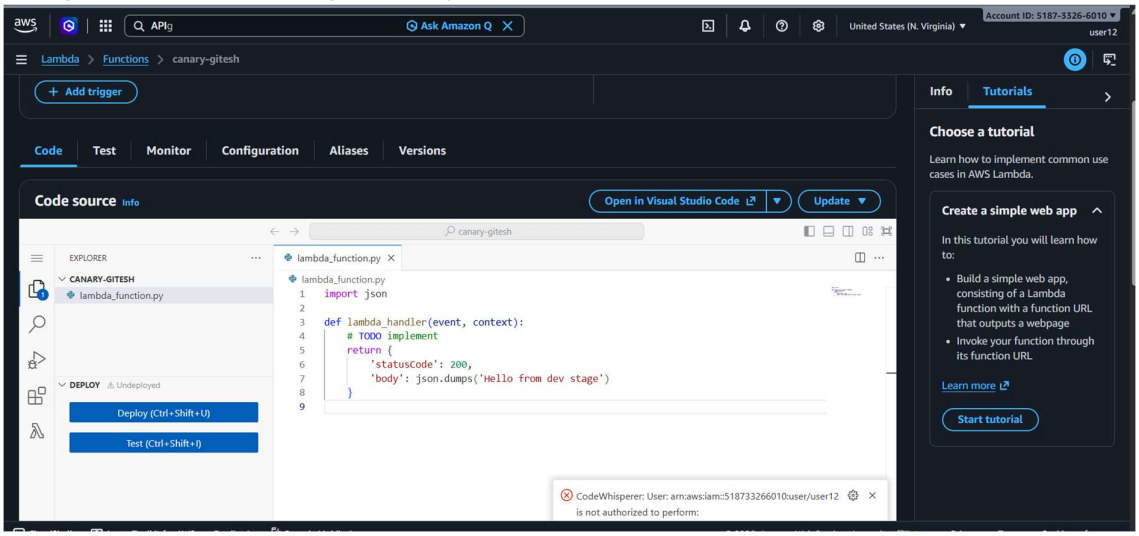
22. Test the URL of API gateway



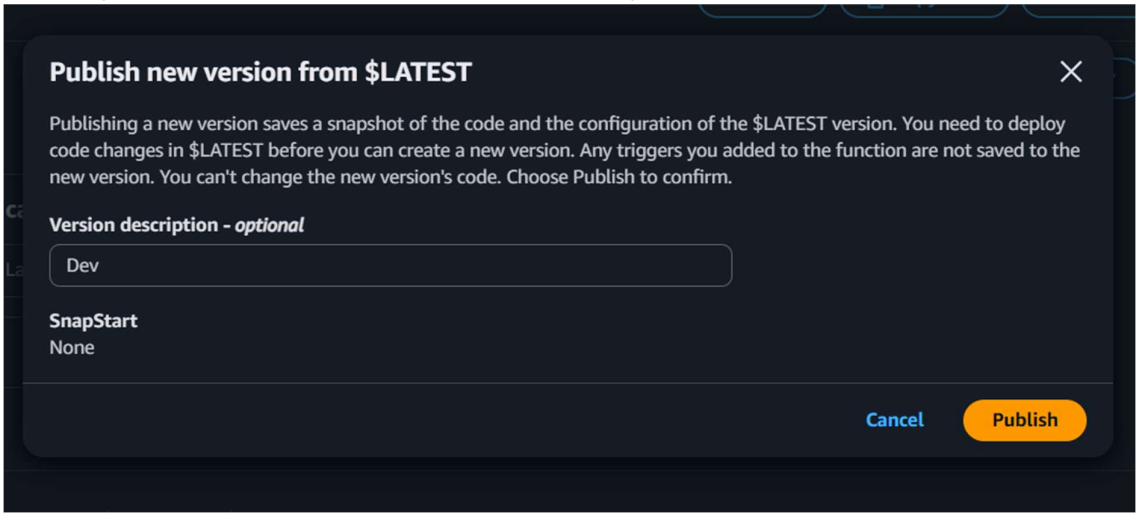
23. At the ned of URL add /getdata as path like this- https://6bg50cj4f6.execute-api.us-east1 .amazonaws.com/prod/getdata
24. And the output should be as follows — "Hello from prod stage"



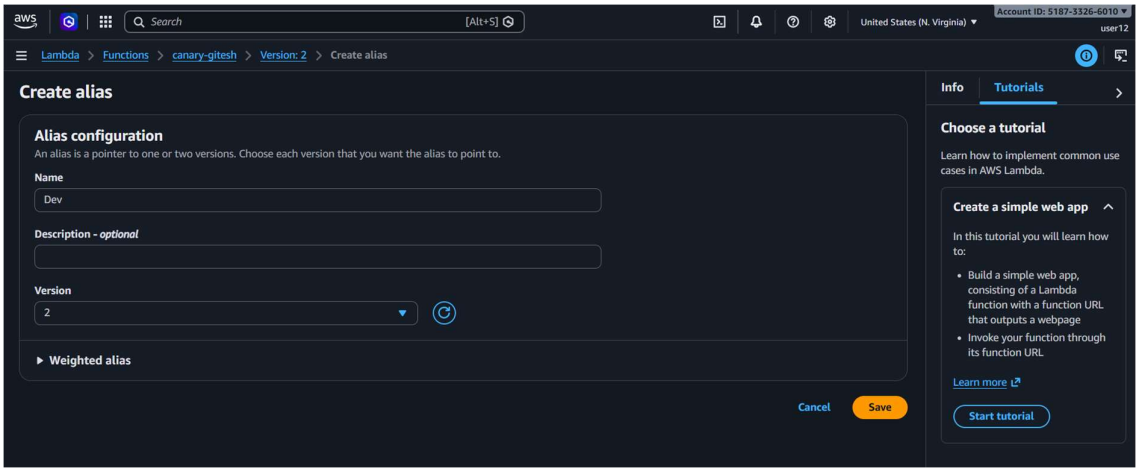
25. Now go to Lambda console again modify the code of $latest version



26. Go to action again, select publish a new version
27. In Description enter name ofversion as Dev and click on publish



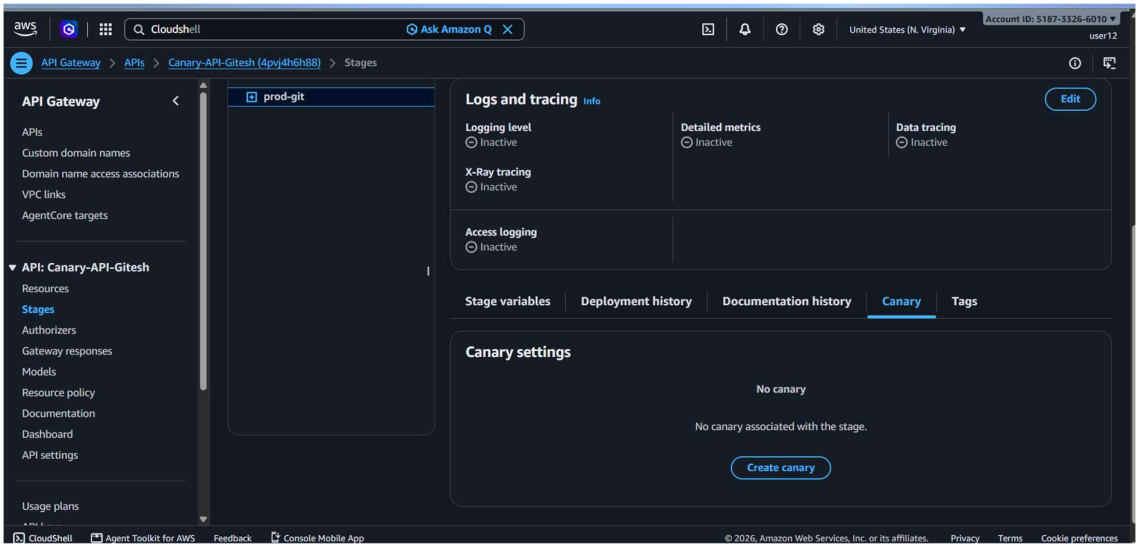
28. Now GO to action again and select create alias
29. Under Create alias window, enter name of alias as Dev select version as 2 and save.



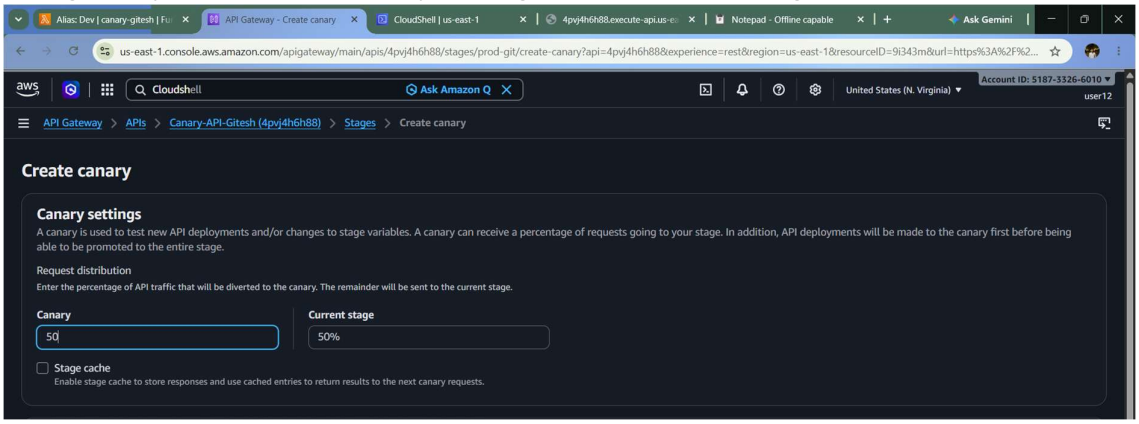
30. Now again we have to add permission using AWS CLI command executed in Task 2 Step no 13
31. This time replace name of version as Dev as follows-
aws lambda add-permission --function-name "arn:aws:lambda:us-east-1:518733266010:function:canary-gitesh:Dev" --source-arn "arn:aws:execute-api:us-east-1:518733266010:4pvj4h6h88/*/GET/getdata" --principal apigateway.amazonaws.com --statement-id 1dd0d748-a5bd-4fd5-9839-2e01c10afc92 --action lambda:InvokeFunction



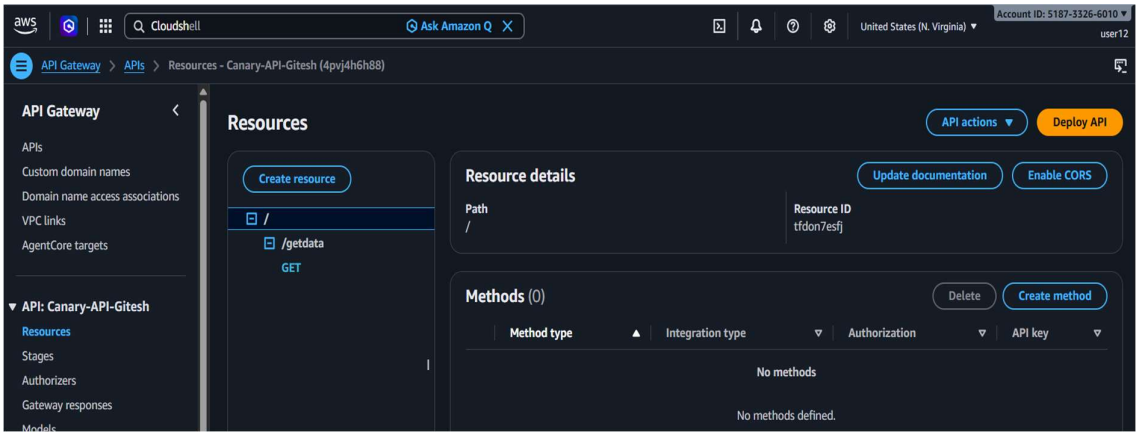
32. Now got to stages and select prod stage and in that select canary tab



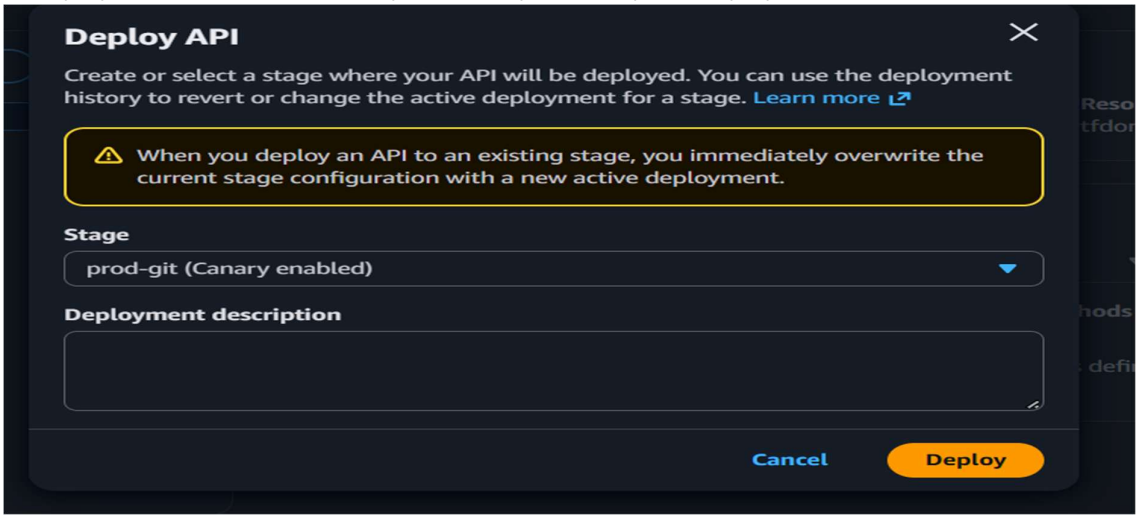
33. Select create canary
34. In Stage's Request Distribution enter percentage as below and select right tick



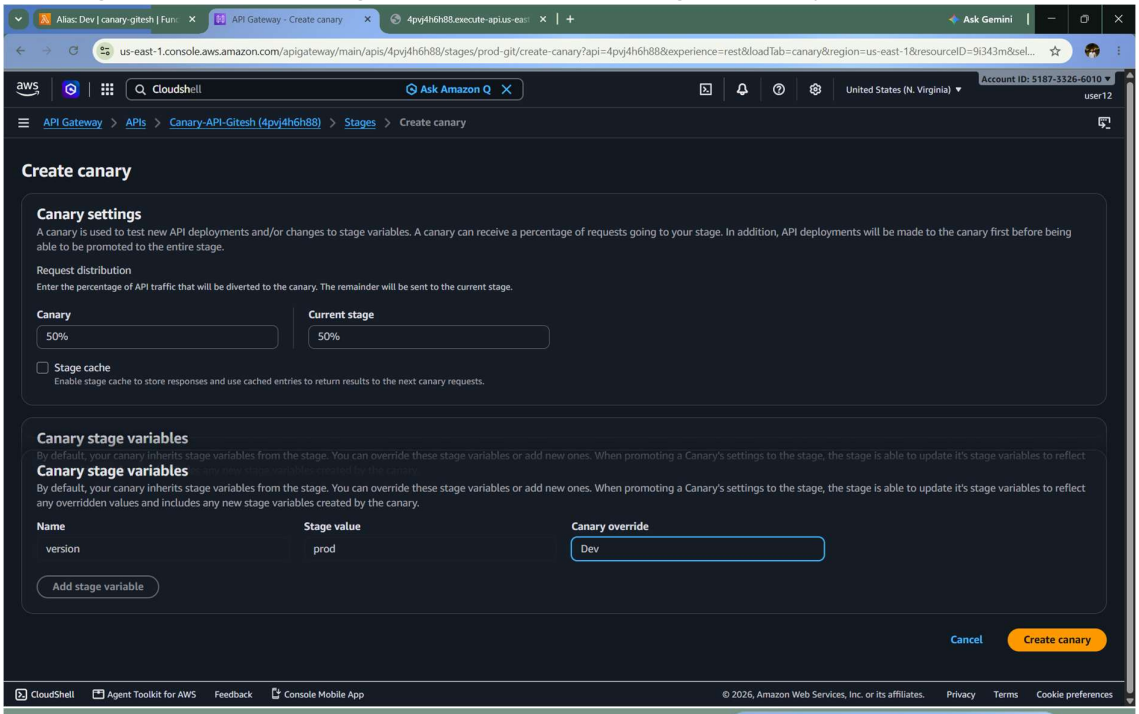
35. Go to Resources and reploy the API again.
36. Go to action and select deploy API.

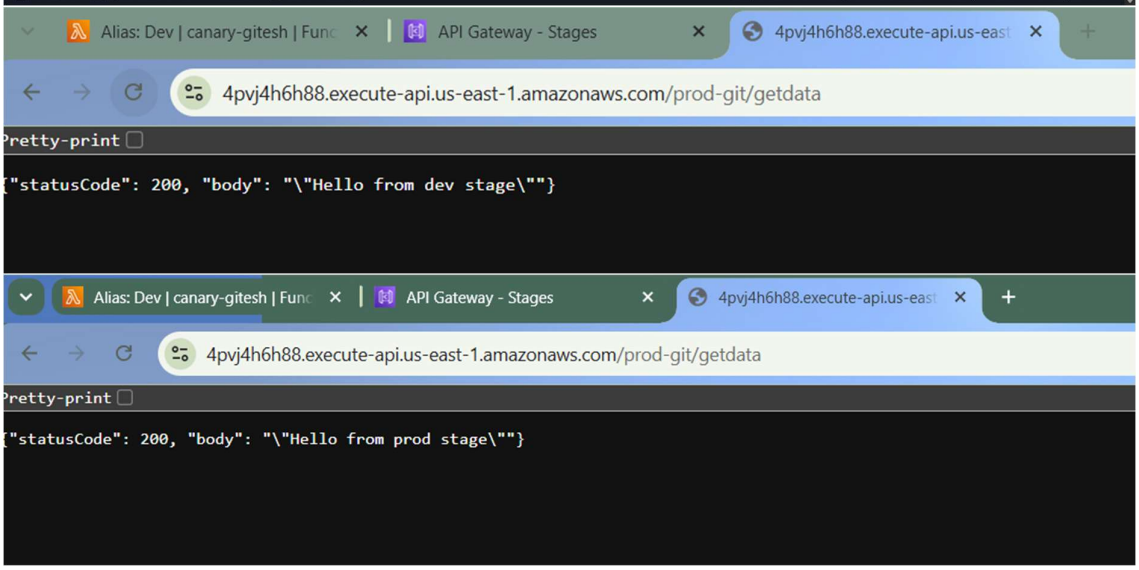


37. In Deploy API window select canary enabled option and press Deploy.

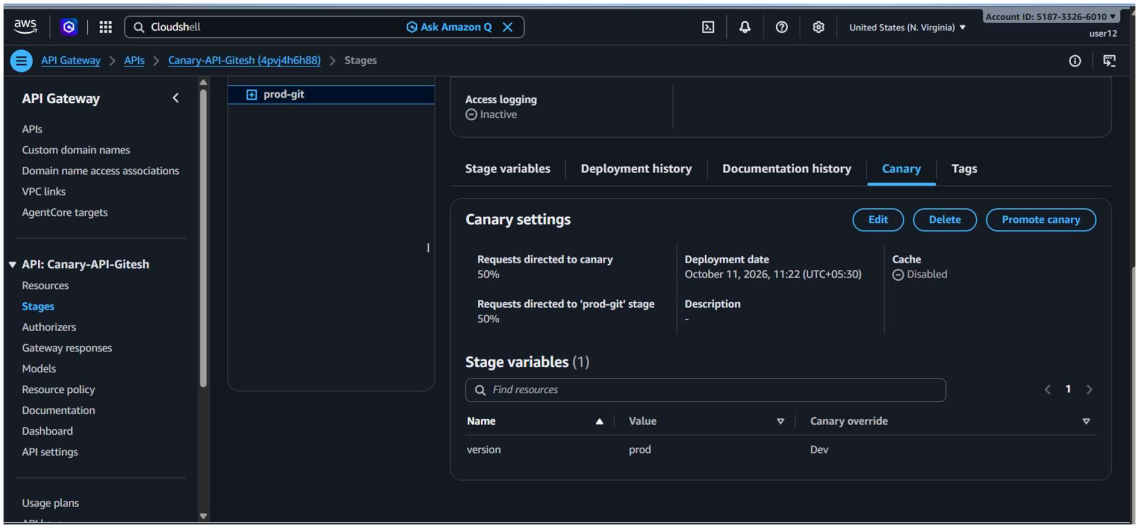


38. Now again Stages under canary tab , we have Canary Stage Variables, edit the cacnary override value with Dev alias now and select right tick.
39. Now invoke the API URL again by appending /getdata path
40. You will get "Hello from Dev stage" and "Hello from Prod stage" randomly.

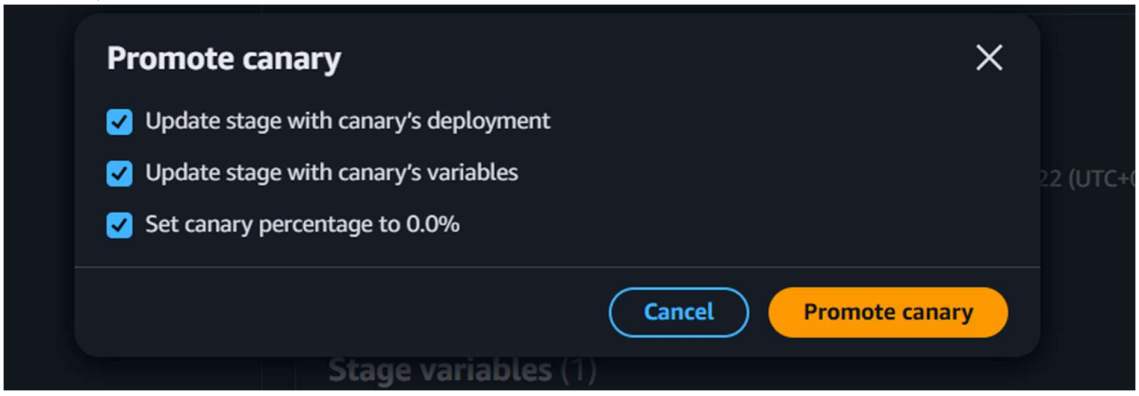




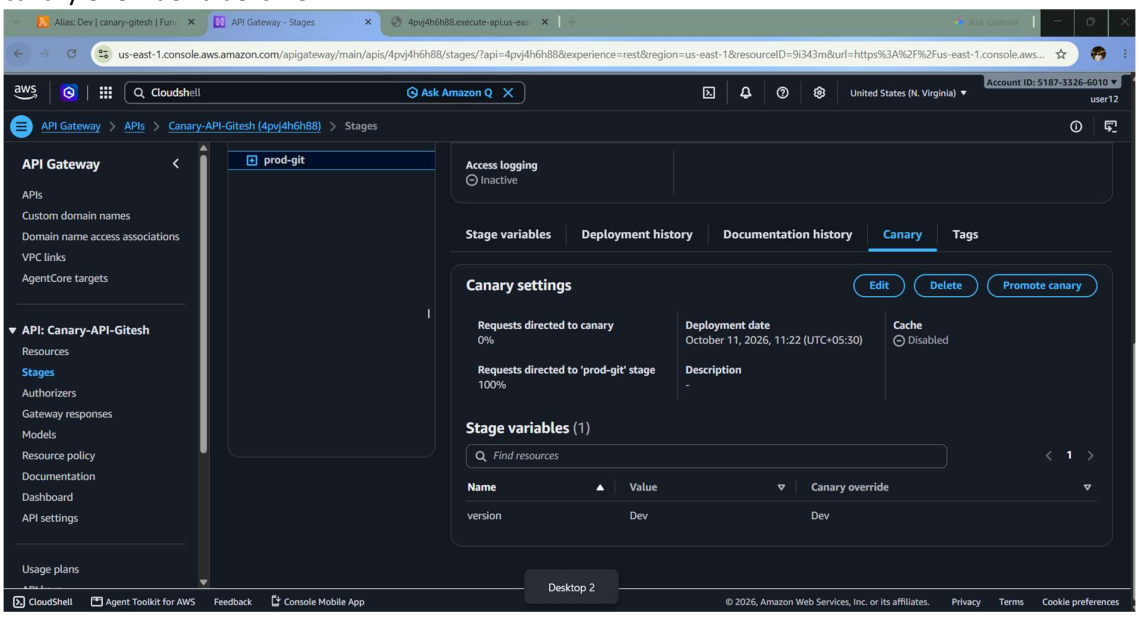
41. Now under canary tab, select promote canary



42. Under promote canary
43. Select Update



44. Once you promote canary you can notice under canary stage variables both stage value and canary Override value is Dev.



45. Now if you hit the API URL again then you will get "Hello from Dev stage" every time.

